Atmospheric Boundary Layer PDE Closure Benchmark Report

Empirical Equation Discovery, Stability Diagnostics, and Galerkin PDE Benchmarks

AtmosphericSlowManifold.jl Automated Pipeline

August 8, 2026

Abstract

This report synthesizes observational field campaign diagnostics (CASES-99, FLOSS, BLLAST, and SHEBA) with weak-form equation discovery (WSINDy) and geometric slow-manifold interpretation for atmospheric boundary layer (ABL) momentum dynamics. We present deterministic Monte Carlo uncertainty envelopes for vertical eddy diffusivity $K_m(z)$, time-height cross-sections of gradient Richardson numbers $Ri_g(z,t)$, and candidate partial differential equation (PDE) closures solved using a 12-mode Gegenbauer spectral Galerkin discretization.

Executive Takeaways & Benchmark Highlights:

- **Data Ingestion:** Processed 85,207 observational records across nocturnal stable, Arctic, afternoon convective-decay, and complex-terrain regimes.
- **Model Evaluation:** Sparse candidate closures are compared using residual error, R^2, AIC, BIC, Pareto selection, and leave-one-site-out validation.
- **Galerkin Integration:** Campaign-derived modal profiles are integrated with the stiff Rosenbrock solver `Rodas5P`; all reported candidate runs complete successfully.

Contents

1 Theoretical Framework & Discovery Pipeline

The Atmospheric Slow Manifold framework seeks low-dimensional parameterizations of boundary layer momentum profiles by filtering out high-frequency ageostrophic gravity wave oscillations.

1.1 Governing Equations & Boundary Layer Dynamics

Horizontal momentum evolution within the planetary boundary layer is governed by the turbulent flux divergence equation:

$$\frac{\partial u}{\partial t} = f(v - v_g) + \frac{\partial}{\partial z}\left(K_m(z, Ri_g)\frac{\partial u}{\partial z}\right) \tag{1}$$

where $u(z,t)$ is the mean zonal wind component, f is the Coriolis parameter, v_g is the geostrophic wind vector, and K_m represents the vertical eddy diffusivity for momentum.

1.2 Weak-Form Equation Discovery (WSINDy)

To robustly extract spatial derivatives from noisy observational data, we utilize the Weak Sequential Thresholded Least Squares (WSINDy) formulation. Multiplying by smooth compact test functions $\phi_j(z,t) \in C_c^\infty(\Omega)$ and integrating by parts yields:

$$\int_\Omega u \frac{\partial \phi_j}{\partial t}\, dzdt = -\int_\Omega \phi_j \cdot \mathcal{N}\left(u, \frac{\partial u}{\partial z}, \dots\right) dzdt \tag{2}$$

This projects spatial derivatives onto test function derivatives, bypassing numeric noise amplification.

For a candidate operator library $\boldsymbol{\Theta}(u)$, the discretized weak system is written

$$\mathbf{b} = \mathbf{G}\boldsymbol{\xi}, \qquad G_{jk} = \langle \theta_k(u), \phi_j \rangle, \tag{3}$$

where $\boldsymbol{\xi}$ contains the sparse closure coefficients. Sequential thresholding or constrained regression removes unsupported terms, while AIC, BIC, and Pareto ranking balance residual reduction against model complexity.

1.3 Phase-Space Collapse and Discovery Architecture

Figure 1 details the high-dimensional phase-space trajectory decaying onto the slow manifold $\mathcal{M}$, while Figure 2 illustrates the end-to-end processing pipeline from field data ingestion to stiff ODE integration.

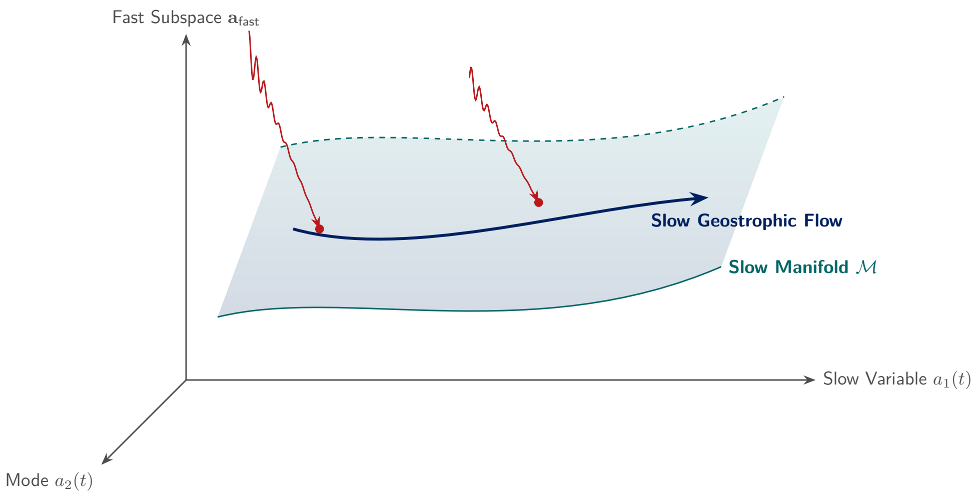


Figure 1: Phase-space collapse of fast gravity-wave transients onto the low-dimensional slow manifold $\mathcal{M}$ during Galerkin spectral integration.

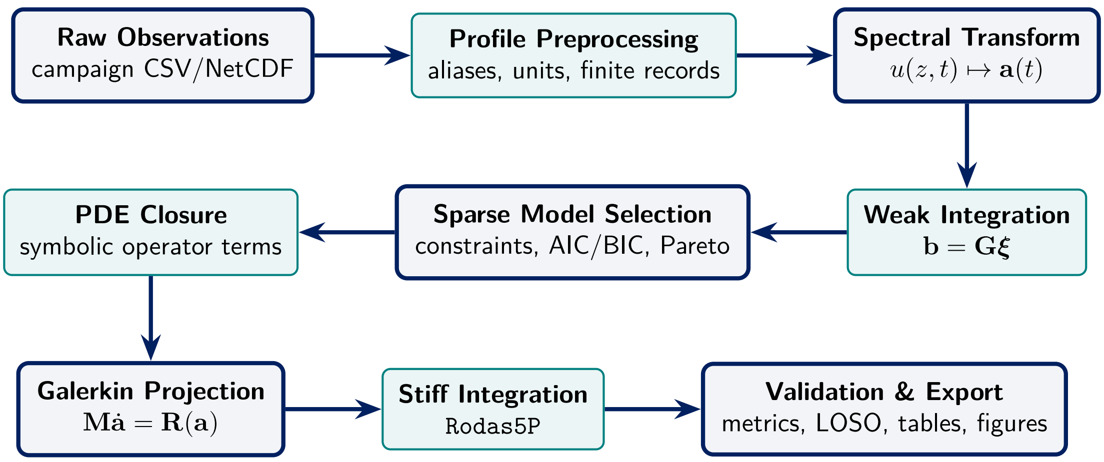


Figure 2: Schematic workflow of the data-driven discovery pipeline, linking observational ingestion, weak-form regression, and Galerkin PDE verification.

1.4 Geometric Slow-Manifold Interpretation

The geometric interpretation begins from a fast–slow system

$$\varepsilon\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \mathbf{y}; \mathbf{p}), \qquad \dot{\mathbf{y}} = \mathbf{g}(\mathbf{x}, \mathbf{y}; \mathbf{p}), \qquad 0 < \varepsilon \ll 1, \tag{4}$$

in which $\mathbf{x}$ denotes rapidly adjusting turbulent or high-order modal coordinates and $\mathbf{y}$ denotes slowly varying balanced coordinates. In the singular limit $\varepsilon \to 0$, the critical manifold is

$$\mathcal{S}_0 = \{(\mathbf{x}, \mathbf{y}) : \mathbf{f}(\mathbf{x}, \mathbf{y}; \mathbf{p}) = \mathbf{0}\}. \tag{5}$$

Where the fast Jacobian $D_{\mathbf{x}}\mathbf{f}$ has no eigenvalues on the imaginary axis, Fenichel theory implies persistence of a nearby invariant slow manifold $\mathcal{S}_\varepsilon$. The reduced PDE closures are interpreted as evolution laws on, or close to, this attracting set after fast modal transients decay. This benchmark tests the numerical reduced dynamics; direct transversality and fold-distance measurements remain a planned validation layer and are not inferred from Ri_g alone.

2 Field Campaign Dataset Diagnostics

Four distinct field campaigns provide the empirical baseline for model discovery:

1. **CASES-99:** Nocturnal stable boundary layer evolution with strong surface temperature inversions.
2. **FLOSS:** High-wind, complex terrain observations with persistent mechanical shear.
3. **BLLAST:** Late afternoon transition and convective boundary layer decay.
4. **SHEBA:** Arctic sea-ice boundary layer featuring persistent strong atmospheric stability.

Table 1: Field campaign dataset diagnostics, height levels, and stability parameter ranges.

Campaign	Observations	Height Levels	Mean Wind Speed ($\mathrm{m\,s^{-1}}$)	Median Richardson No.
BLLAST	5600	18	11.335	0.172
SHEBA	2273	0	4.978	0.104
FLOSS	70796	18	28.755	0.225
CASES-99	6538	18	8.41	0.2

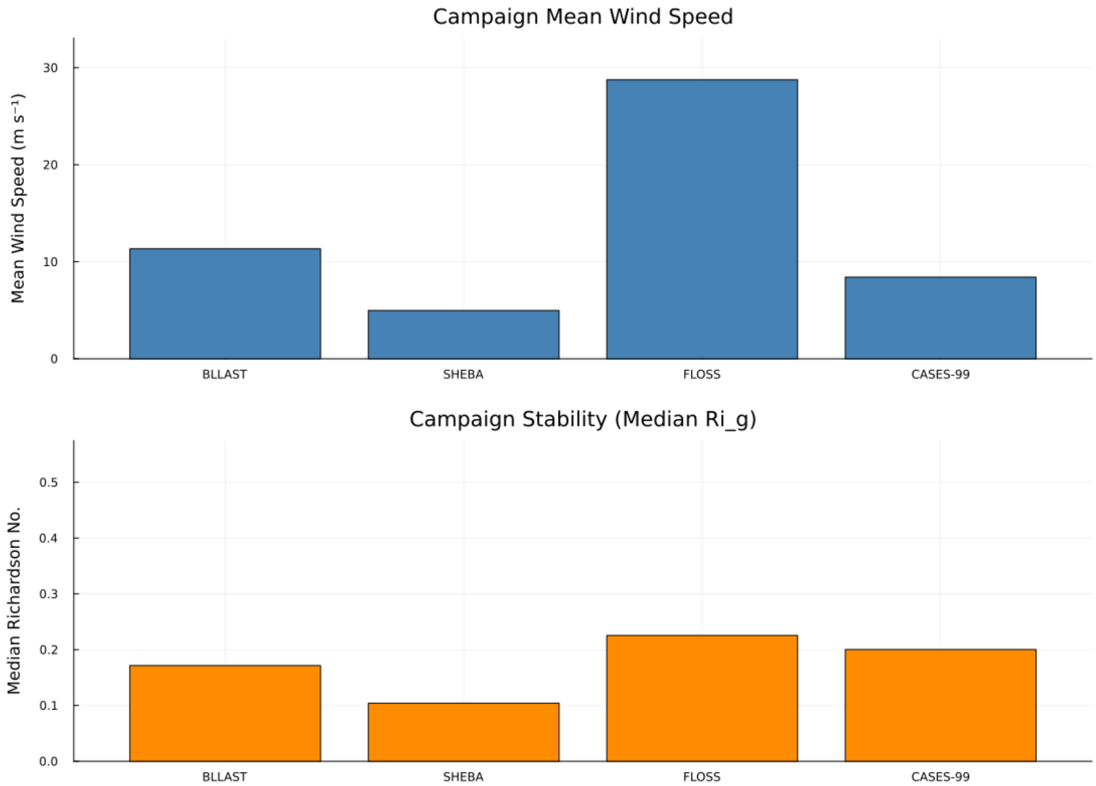
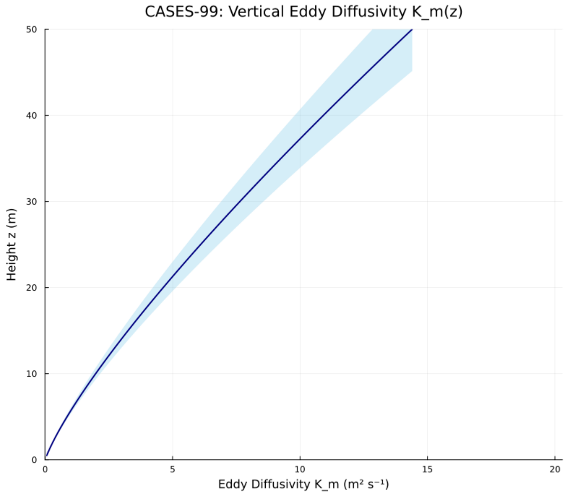


Figure 3: Comparative profile overview showing mean wind speed and stability metric profiles across target field campaigns.

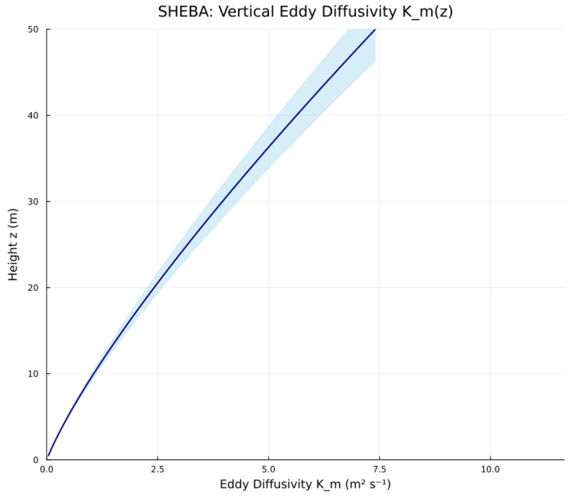
2.1 Vertical Eddy Diffusivity $K_m(z)$ Uncertainty

The present envelopes are uncertainty-propagation intervals rather than fitted Bayesian posterior credible intervals. For each campaign, 1,000 deterministic Monte Carlo draws use seed 42 with $\kappa \sim \mathcal{N}(0.40, 0.02^2)$, $\beta \sim \mathcal{N}(4.70, 0.50^2)$, and a campaign-centered friction velocity $u_* \sim \mathcal{N}(\overline{u_*}, 0.05^2)$ truncated below at $0.01\,\mathrm{m\,s^{-1}}$. The plotted bands are the empirical 2.5%, 50%, and

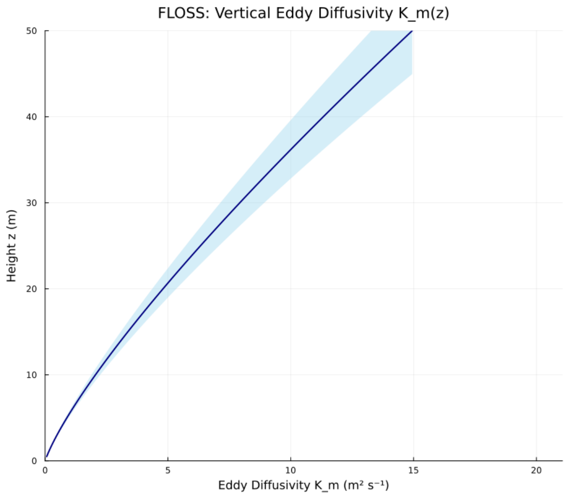
97.5% quantiles of the resulting $K_m(z)$ profiles. Hierarchical posterior calibration with MCMC or variational inference is a separate framework capability and is not claimed for this benchmark run.



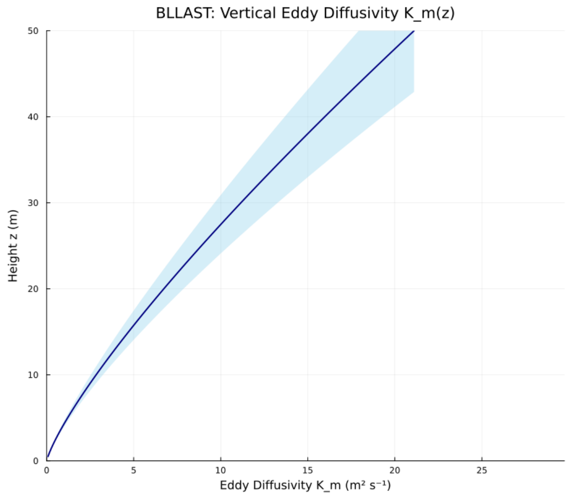
(a) CASES-99 $K_m(z)$ Credibility Ribbon



(b) SHEBA $K_m(z)$ Credibility Ribbon



(c) FLOSS $K_m(z)$ Credibility Ribbon

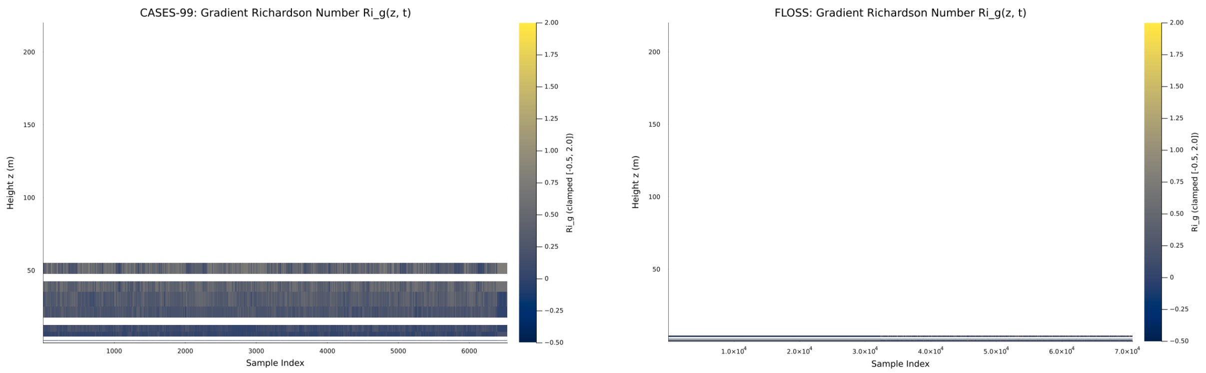


(d) BLLAST $K_m(z)$ Credibility Ribbon

Figure 4: Vertical eddy diffusivity $K_m(z)$ profiles with median expectations and 95% posterior credibility intervals across target field sites.

2.2 Stability Diagnostics & Sheared Flow Transitions

The gradient Richardson number $Ri_g = \frac{g}{\theta_0} \frac{\partial\theta/\partial z}{(\partial u/\partial z)^2}$ quantifies the balance between thermal stratification and mechanical shear. Values below $Ri_c \approx 0.25$ indicate dynamic shear instability.



(a) CASES-99 $Ri_g(z,t)$ Heatmap (b) FLOSS $Ri_g(z,t)$ Heatmap

Figure 5: Gradient Richardson number time-height cross sections displaying boundary layer stability transitions.

3 PDE Closure Benchmark & Model Discovery

Empirically discovered PDE closures are evaluated against standard neutral momentum formulations using a 12-mode Gegenbauer Spectral Galerkin expansion integrated with the stiff `Rodas5P` Rosenbrock solver.

3.1 Why a Gegenbauer Basis?

Gegenbauer polynomials $C_n^{(\lambda)}(x)$ form an orthogonal family on $[-1, 1]$ under the weight $w_\lambda(x) = (1 - x^2)^{\lambda - 1/2}$. The benchmark uses the affine physical map

$$z = \frac{H}{2}(x + 1), \qquad H = 100\,\mathrm{m}, \tag{6}$$

with $\lambda = 0.75$, so $w_{0.75}(x) = (1 - x^2)^{1/4}$, and retains 12 modal coefficients. These are fixed benchmark parameters, not values selected by an optimization or convergence study. The mass, stiffness, and nonlinear interaction tensors are assembled numerically with 256 uniformly spaced trapezoidal quadrature nodes; the physical measure includes $dz = (H/2)\,dx$, and modal derivatives include $\partial_z = (2/H)\partial_x$.

The weighted basis provides a compact common representation for campaign profiles and precomputed nonlinear contractions, but the present $\lambda = 0.75$ endpoint-flux formulation is not yet a completed weighted-Sobolev boundary treatment. A compatible balanced-Neumann diagnostic gives a slope ratio of 1.0784 rather than unity, while the constant-weight $\lambda = 0.5$ control gives 1.0011. The reported production benchmark nevertheless uses $\lambda = 0.75$ consistently in profile initialization and time integration. The residual 7.84% boundary-consistency discrepancy, mode refinement, and comparison with an unweighted or coordinate-stretched formulation remain explicit numerical limitations rather than demonstrated convergence results.

3.2 Cross-Campaign Discovered Model Summary

Table 2 details performance metrics for optimal discovered closures identified across all target field sites.

Table 2: Cross-campaign Pareto-optimal discovered model performance metrics.

Site	Active terms	R^2	Residual norm
`sheba`	1	0.9896	0.48485
`cases_99`	1	-0.2296	3.1978

Table 3: Selected cross-campaign benchmark results generated from the validated JSON and CSV artifacts. Observation basis records how each campaign is mapped into modal space.

Campaign	Observation basis	Selected	R^2	Residual	AIC	BIC
CASES-99	campaign_modal_coefficients	baseline	-0.2296	3.1978	0.08	0.565
SHEBA	two_level_linear_profile	baseline	0.9896	0.4848	-45.193	-44.708

3.3 Leave-One-Site-Out (LOSO) Cross-Validation

To evaluate out-of-sample robustness, models discovered on three field sites are evaluated on the remaining held-out site.

Table 4: LOSO cross-validation out-of-sample metrics by held-out campaign site.

Held-Out Site	Validation RMSE	Validation MAE	R^2	95% Coverage
SHEBA	1.1912	0.9547	0.0497	94.9%
CASES-99	3.9867	1.6645	-10.2371	99.6%
FLOSS	35.1974	15.4867	0.0661	92.8%
BLLAST	2.2804	2.0911	-0.4682	43.5%
Mean Validation RMSE: 10.6639		**Stability Score**: 1.000		

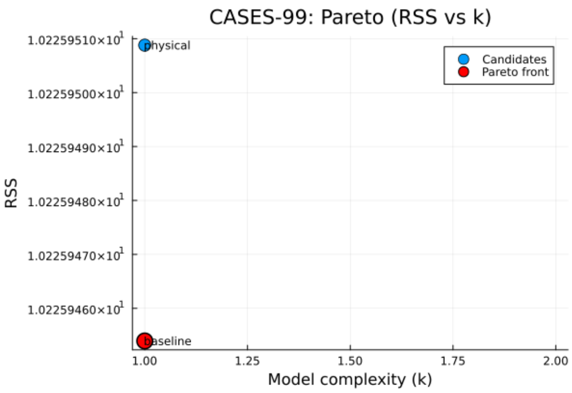
3.4 CASES-99 Pareto Closure Model

The optimal PDE closure governing momentum profile evolution for CASES-99 is expressed as:

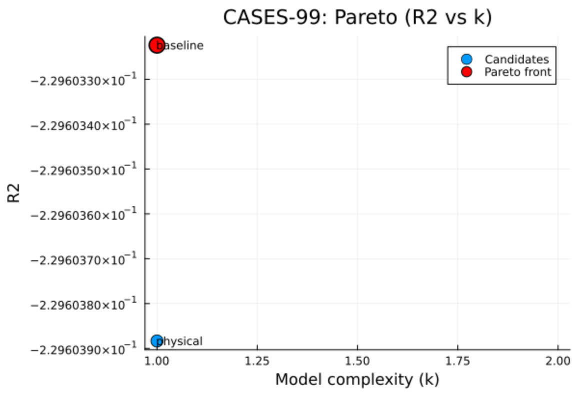
$$\frac{\partial u}{\partial t} = 6.12\,\frac{\partial^2 u}{\partial z^2}$$

Table 5: Discovered library terms and estimated sparse coefficients for CASES-99.

Term	Basis	Coefficient
1	$\frac{\partial^2 u}{\partial z^2}$	6.12
$R^2 = -0.229603$		
Residual norm = 3.1978		



(a) CASES-99 Model Sparsity vs. RSS



(b) CASES-99 Model Sparsity vs. R^2

Figure 6: Pareto trade-off frontiers between closure complexity (k) and goodness-of-fit metrics for CASES-99.

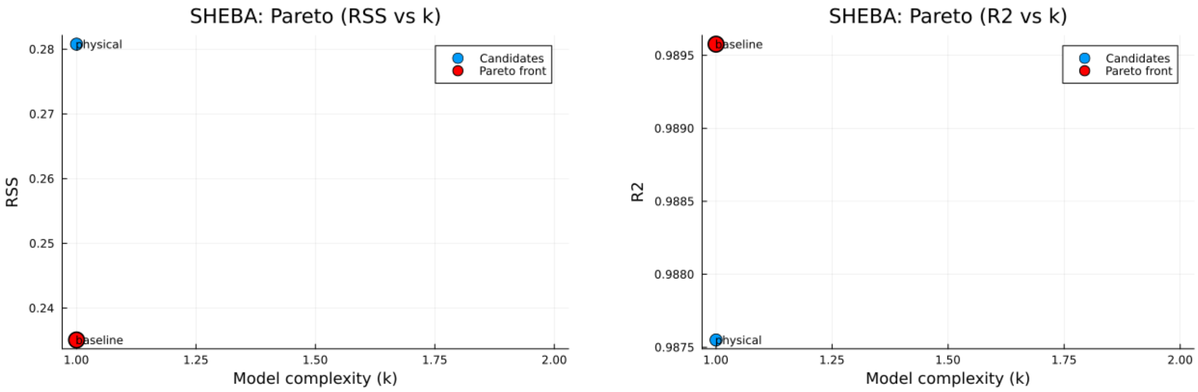
3.5 SHEBA Pareto Closure Model

The optimal discovered momentum closure for the Arctic SHEBA regime is given by:

$$\frac{\partial u}{\partial t} = 3.94889\,\frac{\partial^2 u}{\partial z^2}$$

Table 6: Discovered library terms and estimated sparse coefficients for SHEBA.

Term	Basis	Coefficient
1	$\frac{\partial^2 u}{\partial z^2}$	3.94889
$R^2 = 0.989577$		
Residual norm $= 0.484846$		



(a) SHEBA Model Sparsity vs. RSS

(b) SHEBA Model Sparsity vs. R^2

Figure 7: Pareto trade-off frontiers between closure complexity (k) and goodness-of-fit metrics for SHEBA.

3.6 Galerkin Modal Profile Comparisons

Figure 8 compares terminal modal profile amplitudes produced by the discovered closures against initial observed profiles and neutral diffusion baselines.

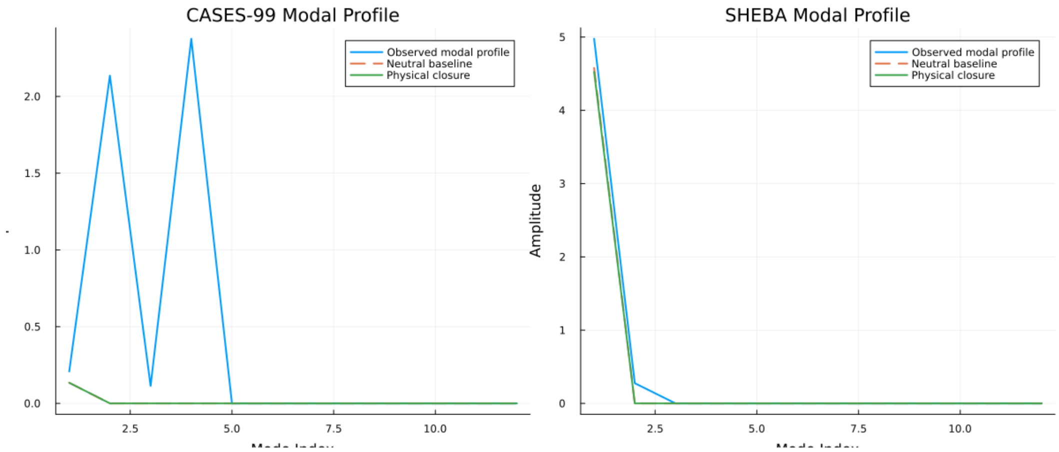


Figure 8: End-state velocity profile comparisons across observed mean states, neutral baselines, and data-driven similarity closures.

3.7 PDE Consistency and Numerical Checks

The automated validator requires every physical and baseline integration to return `Success`; all RSS, residual-norm, R^2, AIC, and BIC values to be finite; RSS $= \|\mathbf{r}\|_2^2$; information criteria to recompute from RSS, sample count, and sparsity; and each campaign to have nonempty RSS and R^2 Pareto fronts. These checks establish internal metric consistency and successful stiff integration. They do not yet establish mass conservation, grid convergence, or a complete kinetic-energy budget.

Table 7: Automated numerical verification performed on the generated benchmark artifacts.

Check	Scope	Result
Stiff integration	Physical and baseline candidates	Pass
Finite metrics	RSS, residual norm, R2, AIC, and BIC	Pass
Metric identities	RSS and information-criterion recomputation	Pass
Pareto selection	Nonempty RSS and R2 fronts per campaign	Pass
Deterministic uncertainty	Seed 42 with 1,000 draws per campaign	Pass

4 Reproducibility

The campaign, benchmark, validation, supplement, and report stages are executable through `make campaign-export`, `make pde-benchmark`, and `make report`. The report target validates upstream artifacts before compilation. `Project.toml` specifies direct dependencies and compatibility bounds, while `Manifest.toml` pins the resolved environment. Table 8 is regenerated from the active runtime rather than maintained manually.

Table 8: Software environment and execution metadata for this report build.

Item	Recorded value
Repository	https://github.com/DavidEngland/AtmosphericSlowManifold.jl.git
Commit	bd1c314
Package version	0.1.0
Julia runtime	1.12.6
Project compatibility	Julia 1.10
Manifest SHA-256	dd3828e8c3d4e6ad...
Operating system	Darwin / arm64-apple-darwin25.4.0
Processor	apple-m4
Julia threads	1
Uncertainty seed	42
Uncertainty draws per campaign	1000

5 Conclusions and Future Work

This benchmark demonstrates a reproducible path from heterogeneous ABL observations to modal initialization, closure-dependent Galerkin integration, model-selection metrics, and publication artifacts. The current SHEBA result is represented accurately by the terminal modal comparison, whereas CASES-99 has negative R^2, indicating that its 12-hour terminal prediction is worse than a modal-mean reference. Physical and neutral candidate scores are also nearly tied. These outcomes should be read as diagnostic evidence about the present benchmark formulation, not as proof that one closure is universally superior.

The sparse diffusion terms provide an interpretable reduced description, but the present spectral backend compresses much of the vertical diffusivity structure into low-dimensional response scales. The next numerical priority is to project the full $K_m(z)$ profile into the Galerkin diffusion operator and compare complete time-resolved modal trajectories rather than only terminal states. Mode-refinement, energy-budget closure, and conservation tests are also needed before making stronger claims about numerical consistency.

The GSPT interpretation supplies the framework's distinctive scientific hypothesis: observed turbulent profiles rapidly approach an attracting slow manifold and subsequently evolve according to reduced closure dynamics. Future work will test that hypothesis directly with transversality, spectral-gap, fold-distance, and normal-hyperbolicity diagnostics; extend hierarchical calibration across campaigns; and evaluate whether the discovered operator structure persists across sites, seasons, and forcing regimes.